ImageNet-R Parent-Recipe Factorial

A same-data causal screen at the one-node task-8, task-16, and task-32 frontiers

> **Outcome.** The strongest cell was Fresh head | wd=5e-4 | joint seed/order. Gap closure was task 16: 100.0%, task 32: 100.0%. The preregistered full-50 trigger was **triggered**.

Abstract

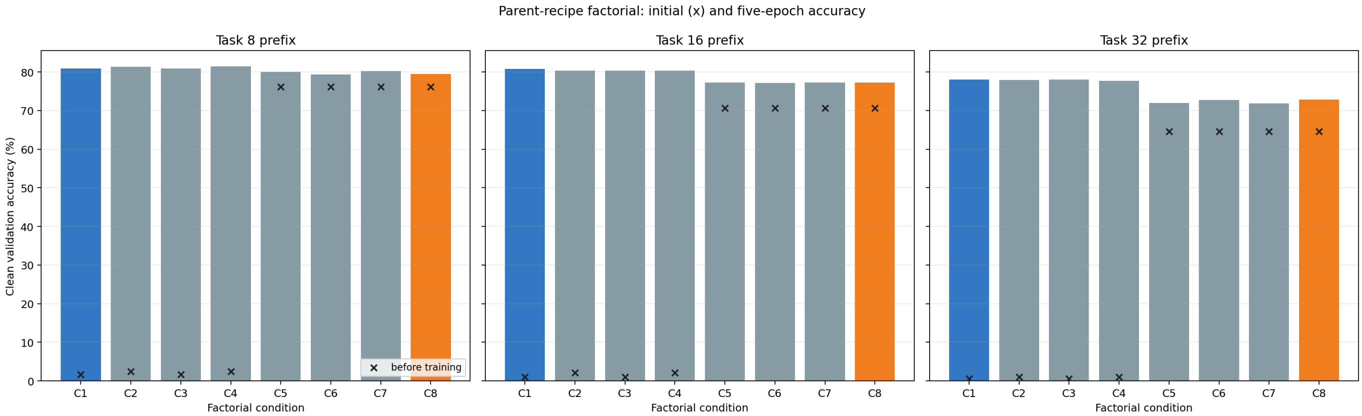
Routing is absent at these power-of-two frontiers, but the consolidated parent lagged a fresh stage-matched joint LoRA. This clean-development experiment computes the complete 2 x 2 x 2 matrix over classifier initialization, weight decay, and seed/data-order schedule. Every cell uses the same pinned ViT-B/16, rank-16 LoRA architecture, examples, and five-epoch budget. No locked-test identity or label is used.

Protocol and exact condition names

The stage-matched joint reference is C1. The exact original full-union parent is C8 and reuses its authenticated source model. The selected cell must close at least half of the C1-minus-C8 gap at both tasks 16 and 32 to trigger a full continual rerun.

Code	Exact condition
C1	Fresh head | wd=5e-4 | joint seed/order
C2	Fresh head | wd=5e-4 | parent seed/order
C3	Fresh head | wd=0 | joint seed/order
C4	Fresh head | wd=0 | parent seed/order
C5	Inherited union head | wd=5e-4 | joint seed/order
C6	Inherited union head | wd=5e-4 | parent seed/order
C7	Inherited union head | wd=0 | joint seed/order
C8	Inherited union head | wd=0 | parent seed/order

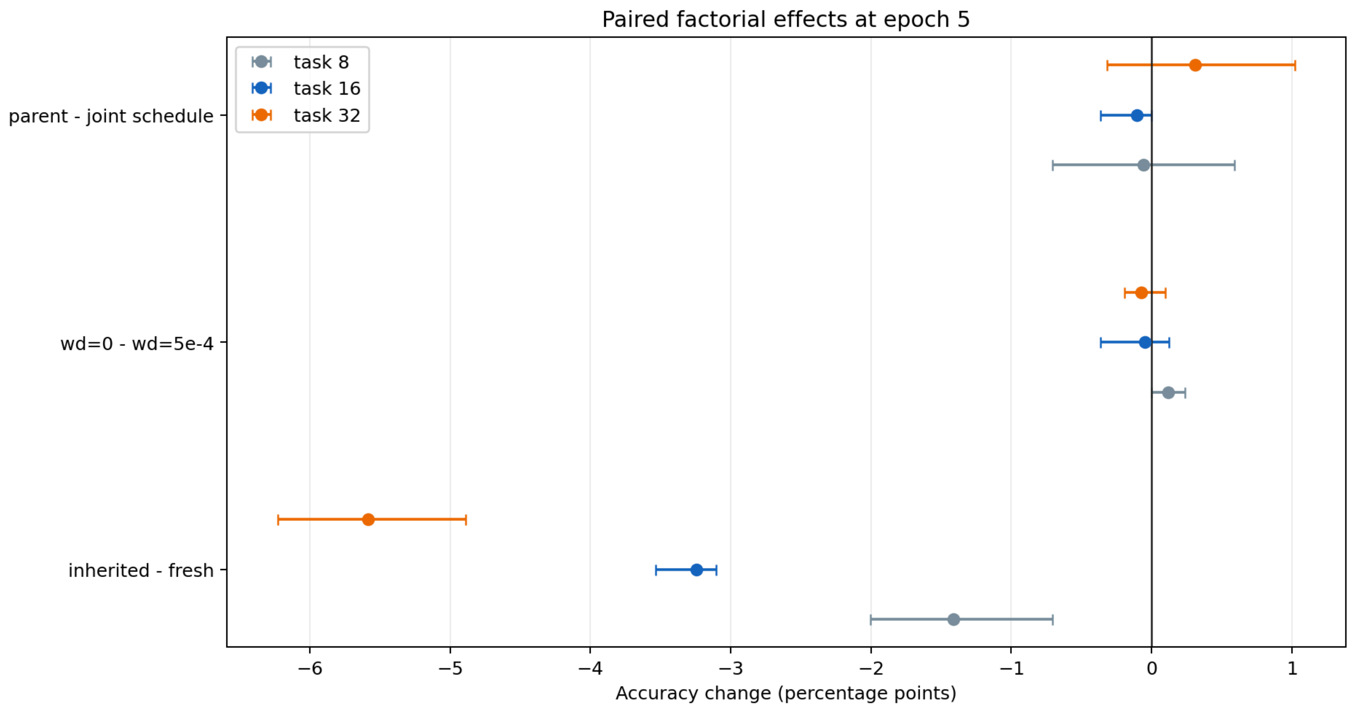
Endpoint results



Crosses are pre-training accuracy; bars are accuracy after five epochs. Blue is the joint-recipe reference and orange is the original parent.

Condition	Task 8	Task 16	Task 32
C1	80.919%	80.755%	78.090%
C2	81.390%	80.390%	77.898%
C3	80.919%	80.390%	78.058%
C4	81.508%	80.390%	77.739%
C5	79.976%	77.223%	72.022%
C6	79.388%	77.162%	72.756%
C7	80.212%	77.284%	71.830%
C8	79.505%	77.284%	72.852%

Factor decomposition



Means and observed ranges across four paired cells, holding the other two factors fixed. These are not confidence intervals.

Stage	Paired contrast	Mean pp	Range pp
8	inherited - fresh	-1.413	[-2.002, -0.707]
8	wd=0 - wd=5e-4	+0.118	[+0.000, +0.236]
8	parent - joint schedule	-0.059	[-0.707, +0.589]
16	inherited - fresh	-3.243	[-3.532, -3.106]
16	wd=0 - wd=5e-4	-0.046	[-0.365, +0.122]
16	parent - joint schedule	-0.107	[-0.365, +0.000]
32	inherited - fresh	-5.581	[-6.228, -4.887]
32	wd=0 - wd=5e-4	-0.072	[-0.192, +0.096]
32	parent - joint schedule	+0.311	[-0.319, +1.022]

Decision

The selected condition is `fresh__wd5e4__joint`. The full-50 trigger was triggered. This trigger allocates compute; it is not an accuracy gate or a statistical-significance claim.

The matrix tests head-state mismatch and optimizer recipe directly. It does not test future-task information, routing, persistent replay, or a longer training budget. The single screening seed meets the 30-minute decision window but does not estimate between-hierarchy uncertainty. Any promoted recipe must therefore be judged by the authorized full continual rerun.

The protocol binds all source roots and children, fit/validation identity hashes, model and dataset manifests, environment, configuration, and material code. Hash-chained per-job ledgers preserve both initial and final validation measurements. CSV, JSON, and Parquet projections are included.